

Designing Synthetic Trigger Peptides via Active Learning Directed Evolution of the SCOOP–MIK2 System

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Background and Research Question

Plant immune systems have evolved sophisticated pattern recognition receptors to detect molecular signals of danger or fungal invasion. One of these is MIK2, a leucine-rich repeat receptor-like kinase. MIK2 is essential for identifying SCOOP peptides, a family of endogenous damage-associated molecular patterns. These 13 amino acid long peptides contain a conserved SxS motif and activate immune responses in *Arabidopsis thaliana* and related species upon recognition by MIK2^[1]. However, the evolutionary success of plant immunity has been tempered by an equally adaptive adversary: the pathogens themselves. Several fungal pathogens, notably *Fusarium oxysporum*, have evolved SCOOP-like peptides that interfere with host signaling. In response, plants have evolved the MIK2 receptor to detect a few of these peptides and trigger defense responses^[2,3,4]. This phenomenon illustrates a molecular arms race, where plants continually refine receptor sensitivity, while pathogens evolve convergent or deceptive molecular patterns that can evade or alter immune recognition.

While nature's evolutionary pace is limited by selection and mutation, computational design offers a faster, more directed means of escaping this stalemate. The core question of this project, therefore, is: Can we computationally design synthetic SCOOP-derived peptides that retain the conserved SxS motif but show improved and wider MIK2 binding as compared to natural analogs? Addressing this question not only sheds light on the structural and evolutionary basis of receptor-ligand specificity but also establishes a generalizable framework for creating synthetic immune primers for crops. The long term goal is to use these designed peptides as broad spectrum immune activators, either as foliar treatments or microbial delivery products. This aims to enhance crop resilience against evolving pathogens while reducing dependence on chemical fungicides^[5].

Methodology and Preliminary Results

Phylogenetic Context: MIK2 Clustering

I began by mapping the evolutionary diversity of MIK2 receptors across Brassicaceae species to find representative receptor homologs for modeling. I constructed a rooted maximum likelihood phylogenetic tree from MIK2 orthologs^[1], from which I derived a patristic distance matrix. Using this matrix, I performed hierarchical clustering^[6] (average linkage) directly on the computed distances. Following this, I applied multidimensional scaling to project the distance matrix into two dimensions while preserving pairwise relationships. This analysis revealed distinct clusters of MIK2 homologs (Fig. 1), with *A. thaliana*, *D. strictus*, *C. violacea*, and *T. cacao* (outgroup) emerging as the centroid cluster representatives. *A. thaliana*, representing the MIK2 clade cluster, was chosen for subsequent peptide modeling. The outgroup was included to provide an evolutionary reference point for rooting and scaling the tree.

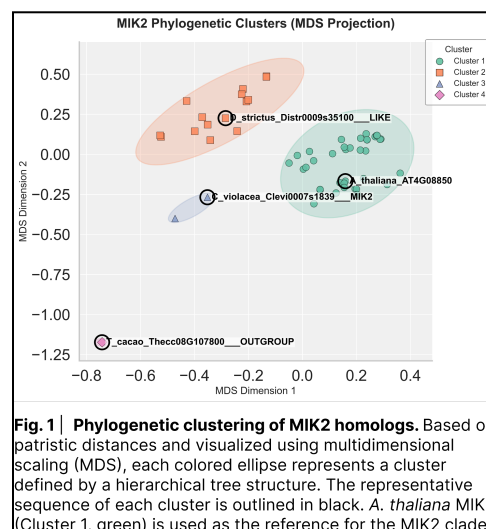


Fig. 1 | Phylogenetic clustering of MIK2 homologs. Based on patristic distances and visualized using multidimensional scaling (MDS), each colored ellipse represents a cluster defined by a hierarchical tree structure. The representative sequence of each cluster is outlined in black. *A. thaliana* MIK2 (Cluster 1, green) is used as the reference for the MIK2 clade.

Directed Peptide Evolution via Active Learning

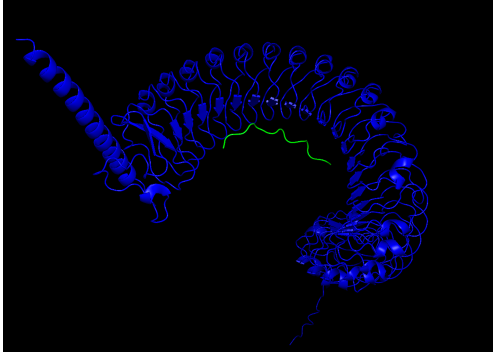


Fig. 2 | Predicted MIK2-SCOOP peptide complex. The MIK2 ectodomain (blue) is bound to the highest fitness SCOOP peptide (green) predicted by ALDE (modeled using AlphaFold3)

To explore peptide sequence space beyond the naturally occurring SCOOPs, I designed a 13-mer SxS constrained peptide domain that combined mutational and *de novo* variants. The conserved serine SxS motif at positions 5 and 7 was fixed to preserve biological function^[1]. The input included 39 known experimentally validated SCOOP and SCOOP-like peptides from *A. thaliana* and *F. oxysporum*^[1,2]. For each seed peptide, I generated 100 mutants by introducing one or two random amino acid substitutions at non-conserved positions, resulting in 3900 variants. To ensure exploration of novel sequences, I generated 8000 additional random peptides *de novo*, maintaining the 13-aa length and the SxS motif. The final domain, comprising ~11500 unique peptides, spanned both near native and novel regions of the sequence space and formed the foundation for our *in silico* directed evolution.

To identify the high fitness variants, I implemented an **Active Learning Directed Evolution (ALDE)** framework^[7]. Each 13-mer peptide was encoded as a 260-dimensional (13×20) one-hot vector. This captured position-specific amino acid information while preserving residue-level variation. I trained an ensemble of deep neural networks on the labeled seed peptides, where the target variable *fitness* corresponded to AlphaFold 3^[8] predicted MIK2 binding confidence (iPTM). The ensemble produced mean and variance predictions across the unlabeled domain, enabling uncertainty-aware Bayesian optimization. New candidate peptides were selected using Thompson Sampling, which probabilistically balances exploration of uncertain regions with exploitation of high-confidence zones. A Hamming distance filter (≥ 3) ensured diversity among selected sequences^[9], resulting in 96 predicted peptides.

I modeled all the 96 ALDE-designed peptides using AlphaFold 3 (Fig. 2) to predict receptor-ligand complex formation and interactions. I installed and ran AlphaFold 3 on the Georgia Tech Phoenix HPC^[10] system within a fully containerized environment for reproducibility. The setup process involved: (i) building the AlphaFold 3 Docker image with Colima, (ii) converting the image into an Apptainer container, (iii) transferring model weights and downloading the necessary databases for AlphaFold 3, and (iv) executing structure prediction jobs for all the peptide variants. Motif analysis of the predicted SCOOP peptides shows that this workflow successfully predicted novel variants that are neither in the training set nor naturally occurring (Fig. 3). These variants conserve the SxS core and feature new amino acids at flexible positions, indicating guided exploration of the sequence space.

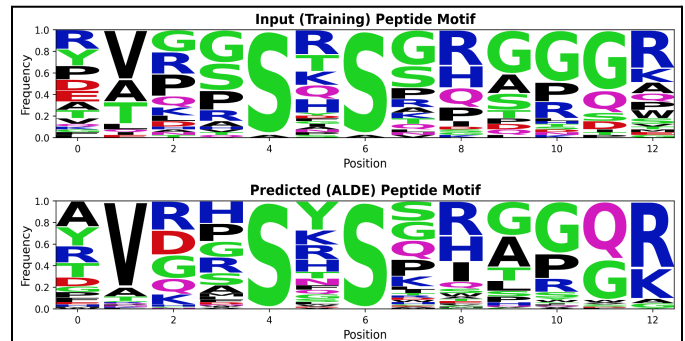


Fig. 3 | Comparison of input and ALDE-designed SCOOP peptide motifs. The conserved SxS motif is retained while ALDE introduces diversity, expanding sequence space and predicting novel variants.

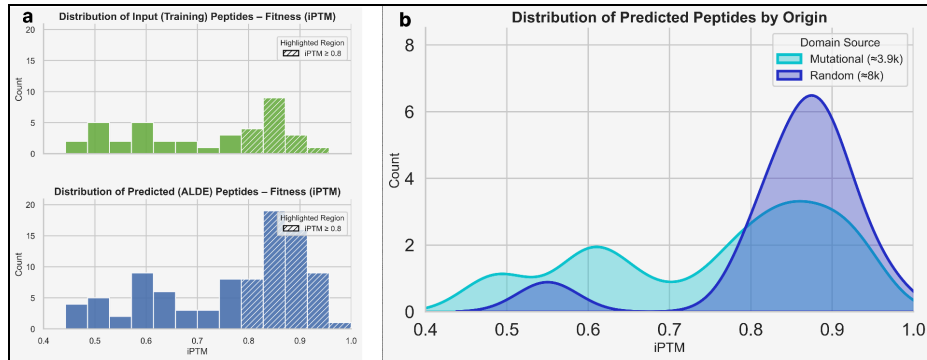


Fig. 4 | Distribution of AlphaFold3 iPTM scores for ALDE predicted peptides.

(a) The training set (green) represents natural SCOOP and SCOOP-like peptides used to initialize the model, while the ALDE distribution (blue) shows newly designed sequences predicted to achieve higher binding scores.

(b) Mutational variants (teal) are local sequences from known SCOOPs, while random peptides (blue) are *de novo* SxS-constrained sequences. Both show high predicted fitness (iPTM ≥ 0.8), with random peptides peaking higher—indicating ALDE's ability to discover novel, high-affinity candidates beyond the native

The AlphaFold 3 output included ranking scores, interface confidence metrics, and the modeled receptor-ligand structures. Preliminary analysis revealed that **~60% of designed peptides achieved iPTM ≥ 0.8 (high confidence protein interaction)**, comparable to or exceeding top natural SCOOPs (Fig. 4a). This demonstrates that the ALDE-guided exploration was capable of discovering novel synthetic binders with high structural confidence (Fig. 4b).

The Plan Ahead

The current phase of this research demonstrates the feasibility of using *in silico* active learning for directed evolution of SCOOP-like peptides with improved MIK2 binding. Moving forward, with the inputs and suggestions from my mentor and principal investigator, Dr. Lily Cheung, I foresee two complementary avenues that will define the next phase of this project. The first will focus on **experimental validation** of the computationally designed peptides. The top candidates generated by the ALDE–AlphaFold 3 pipeline will be synthesized and tested for immune activation in *A. thaliana*. Quantitative assays, including ROS burst, MAPK phosphorylation, and Ca^{2+} influx, can be used to measure receptor activation and compare performance against native SCOOP peptides^[4]. This experimental evidence will not only benchmark the *in silico* predictions but will also refine the learning model by providing new, ground-truth fitness data for further iterative retraining.

The second part will apply **directed evolution to the MIK2 receptor**. Using mutational scanning and AlphaFold 3 structural modeling, I plan to systematically screen and mutate residues within the ligand-binding interface^[11] of the MIK2 ectodomain to identify variants with higher specificity. This can be applied to create genetically modified plants with receptors that can detect pathogen SCOOP-like peptides more effectively than wild-type MIK2, thereby enhancing disease resistance. Another objective is to create **orthogonal SCOOP–MIK2 pairs** that respond exclusively to synthetic SCOOP variants not recognized by native MIK2. These orthogonal circuits could be used in plant synthetic biology to create new signaling pathways without disrupting endogenous systems.

Coming from a Computational Biology background, I have always been passionate about leveraging AI to solve problems with broader, practical applications. I want to use the protein engineering skills that I gain from this project to work towards my passion, drug development. The core ideas and methods used here can be extended to human drug targets, protein interaction inhibitors, or antibody design. I'm confident that this research will enable me to not only analyze biological systems better but also build AI-driven drug discovery pipelines at the forefront of therapeutic development.

Statistical Modeling of Peptide Biosensors using Maximum Likelihood Estimation

My second project focuses on modeling the response of fluorescence biosensors that recognize sugars. The work builds on previously ongoing efforts in high-throughput biosensor design in our lab. These biosensors are derived from the transporter proteins SWEETs, where short linker peptides are present between fluorescent proteins (cpFP) and the transporter to couple conformational change in the transporter to fluorescence output. Because these biosensors are chimeric proteins and consist of only a few amino acids, their behavior cannot be predicted solely from their sequence. Therefore, our lab used a high-throughput sort-seq workflow^[12] where 103,718 linker variants were expressed in yeast, sorted into four logarithmic fluorescence bins using FACS under glucose and starved conditions, and sequenced to obtain read counts per bin.

Traditional bin-averaging approaches are prone to bias because fluorescence distributions are typically log-normal and often asymmetric. To address this, I implemented a **Maximum Likelihood Estimation** framework that models fluorescence intensity in $\log_{10}$ space as either a normal or skew-normal distribution. For each peptide, I calculated the probability mass in each fluorescence bin from the cumulative distribution function of the fitted model, incorporating a small regularization term to avoid zero-probability bins. I used the L-BFGS-B algorithm for optimization for its smooth convergence under explicit parameter bounds. Instead of setting arbitrary limits, I derived adaptive bounds for μ (mean) and σ (standard deviation) using their interquartile range, ensuring realistic fits across peptides with varying expression levels. The expected fluorescence in linear scale was then computed as $10^{(\mu + \sigma^2/2)}$, and the dynamic range was defined as $((\text{mean_glucose} - \text{mean_starved})/\text{mean_starved})$. To account for asymmetry in peptide dynamic ranges, I extended the model with a skew-normal distribution, introducing a skew parameter (α), and applied a Likelihood Ratio Test to assess its significance. I compared the fitted distributions against the experimental histograms to evaluate their accuracy.

This work allows us to better understand the correlation between changes in linker peptide sequence and the corresponding fluorescence response of the biosensor. Moving forward, I plan to use the **Active Learning framework** that I've implemented for the SCOOP peptide synthesis to predict linker sequences. By integrating the Maximum Likelihood Estimated parameters and dynamic range estimates, the model will iteratively propose sequences predicted to improve fluorescence response. These candidates can then be experimentally tested, refit into the statistical model, and added to the training dataset in an active learning feedback cycle. This approach will accelerate the discovery of optimal biosensors while reducing experimental cost, tools that our lab has already demonstrated can be effective in drug discovery.

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